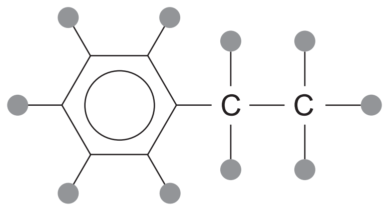
Examiner only

10. (a) Compound **B** contains only carbon, hydrogen and bromine. The hydrogen and bromine atoms could be at any of the positions indicated by a circle in the following structure.



- 5.35 g of compound **B** contains 3.74 g of bromine and 0.11 g of hydrogen
- It rotates the plane of plane polarised light
- When 4.36 g of compound **B** is heated with aqueous sodium hydroxide, the bromide ions produced react with silver ions to give 4.77 g of silver bromide (M_r 187.9)

Use **all** of this information to deduce a possible structure for compound **B**.

Explain your reasoning. [6 QER]

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(b) Sucrose, $C_{12}H_{22}O_{11}$, is a sugar that contains eight —OH groups in each molecule. It reacts with ethanoic anhydride to give sucrose octaethanoate.

sucrose + ethanoic anhydride $\longrightarrow$ sucrose octaethanoate + ethanoic acid

(i) A 0.0700 mol sample of sucrose was used in a preparation.

Calculate the minimum volume of ethanoic anhydride needed to react with all the —OH groups present in this sample of sucrose. [2]

	M_r	Density / g cm^{-3}
ethanoic anhydride	102	1.08

Minimum volume = ... cm^3

(ii) There is interest in producing materials by 'green' methods.

Two methods are proposed for making sucrose octaethanoate in the laboratory.

Method **X** Refluxing the mixture at 139 °C using an electric hotplate

Method **Y** Ultrasonic irradiation at room temperature

Apart from temperature, suggest **two** factors that should be considered when deciding which of these two methods is 'greener'. [2]

1. ....

....

2. ....

....

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(c) Lactose is a reducing sugar and reacts with Fehling's reagent.

(i) State what is seen when lactose reacts with the dark blue solution of Fehling's reagent. [1]

....

(ii) State which functional group must be present in a molecule of lactose for this result with Fehling's reagent. [1]

....

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